# SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

## Project Based Learning – Batch 2023-27

## Title of the project – EMG-Controlled 3D-Printed Prosthetic Hand with Myoelectric Actuators

### INTRODUCTION

This project enhances accessibility in assistive technology by developing a low-cost, 3D-printed, EMG-controlled prosthetic hand. It integrates myoelectric control, real-time sensing, and a modular open-source design to deliver a functional, affordable, and customizable solution for upper-limb loss.

### OBJECTIVES

- To design a shoulder-to-finger 3D-printed upper-limb prosthetic using high-torque servo motors for basic assistive use without clinical fitting.
- To develop a control system using EMG sensor thresholds and an ESP32 to enable simple muscle signal-based operation.
- To evaluate performance on a test rig based on grip responsiveness, control accuracy, signal reliability, and usability.

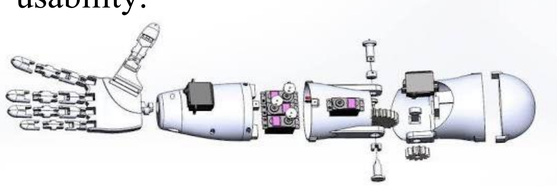


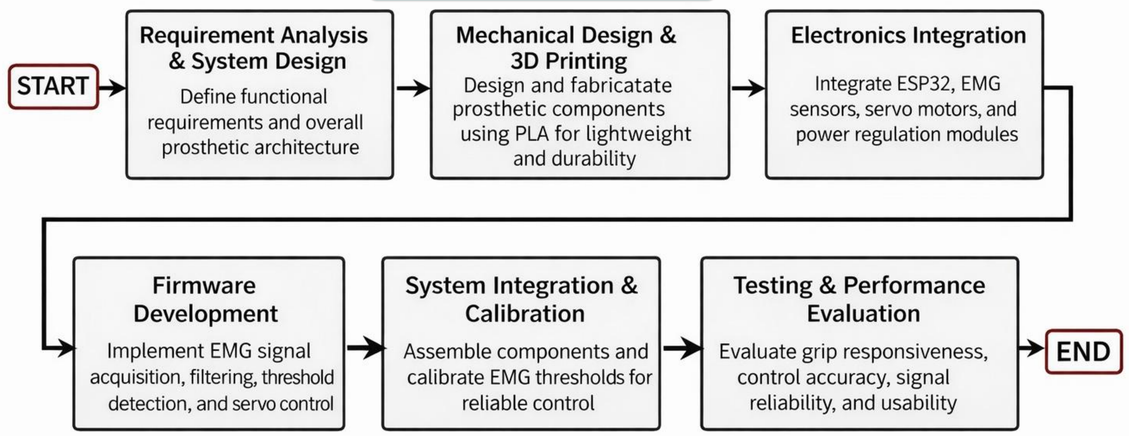
Fig. 1: Initial Design of the Prosthetic Arm

Fig. 2: Fully Assembled 3D-Printed Prosthetic Hand Prototype

### PROBLEM STATEMENT

Current upper-limb prosthetics are either prohibitively expensive or, in academic prototypes, lack integration, reproducibility, and standardized evaluation.
To address this within undergraduate constraints, there is a need to develop and validate a low-cost, modular EMG-controlled prosthetic hand that integrates mechanical design with myoelectric control and is evaluated using measurable performance criteria.

### METHODOLOGY



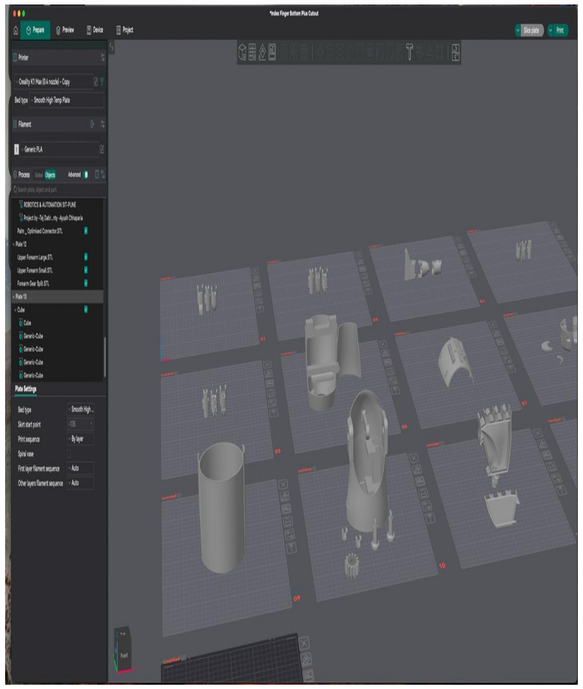



Fig. 3: CAD Workspace Showing Segmented Components for 3D Printing

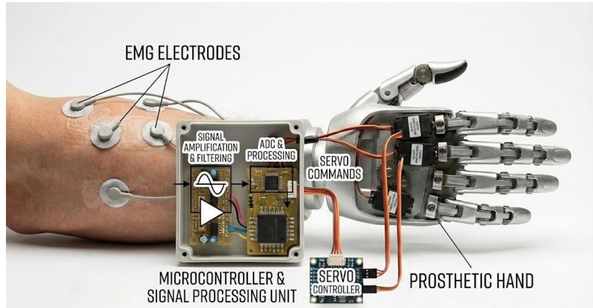



Fig. 4: Internal Electronics Integration with the Prosthetic

Fig. 5: Internal Electronics Integration with the Prosthetic

### RESULTS

- Lightweight 3D-printed, tendon-driven structure.
- Real-time ESP32–EMG control for stable signal processing.
- Responsive, muscle-synchronized grasp via precise servo actuation.
- Low-cost, modular design for research and education.

Fig. 6: Functional Demonstration of Prosthetic Hand Grasping an Object

### CONCLUSION

This project presents an affordable, 3D-printed myoelectric prosthetic hand. ESP32-based EMG control enables reliable signal acquisition and responsive tendon-driven grasping, while the modular design enhances accessibility and supports reproducible research and education.

### NAME OF THE STUDENTS & GUIDE (S)


| Name of the Students | Name of the Guide |
|---|---|
| 1. Priyanshu Mohanty(23070127045) | Dr. Satish Kumar |
| 2. Tej Datir(23070127062) | **Name of Co-Guide** |
| 3. Ayush Chhaparia(23070127015) | Dr. Ramesh B T |
| 4. Shounak Chaudhari(23070127070) | |


## DEPARTMENT OF ROBOTICS AND AUTOMATION